NAMES(S):

Hidden Markov Models and the Viterbi algorithm

To create an HMM, you'll need:

① Set of **states**

{ Hot, Cold } = {H,C}

---

② Sequence of **observations**

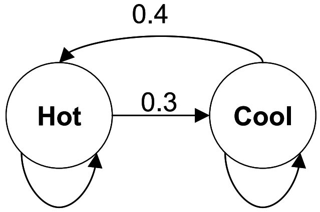
( =

June			
1	***2***	***3***	***4***
3 cones	1 cone	1 cone	…

---

③ Table of **transition probabilities**

Note: Pr (A|B) = probability of event A given that event B has occurred or is true



Pr (H | H) = 0.7
Pr (C | H) = 0.3
Pr (H | C) = 0.4
Pr (C | C) = 0.6

---

④ Table of **emission probabilities**

Pr  | H) = 0.2	Pr  | C) = 0.5
Pr  | H) = 0.4	Pr  | C) = 0.4
Pr  | H) = 0.4	Pr  | C) = 0.1

---

⑤ **Starting probabilities**

Pr (H) = 0.8
Pr (C) = 0.2

Goal: Use the Viterbi algorithm to deduce the most likely sequence of hidden states from the sequence of observations.